MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
SECOND SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)

MG2DSCIAM101 -Automotive Systems for E-Vehicles

Duration: 1.5 hours **Maximum Marks: 50**

Students should attempt atleast one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A
Multiple Choice Questions
Answer **All** Questions
Each question carries 1 mark

1	In a comparator circuit, when the non-inverting input voltage is higher than the inverting input voltage, the output will be: A) High B) Low C) Zero D) Undefined	U	CO1
2	What is the main function of an Operational Amplifier A) Amplify high-frequency signals B) Amplify low-frequency signals C) Perform mathematical operations like addition, subtraction, etc. D) Convert digital signals to analog signals	K	C01
3	The primary purpose of a Unity Gain Amplifier (Buffer Amplifier) using an Op-Amp is: A) To amplify the input signal with high gain B) To provide isolation between different stages of a circuit C) To filter the input signal D) To oscillate at a fixed frequency	U	C01
4	The primary function of a Schmitt Trigger is A) To amplify a signal without distortion B) To convert a sinusoidal signal into a square wave C) To filter high-frequency noise from a signal D) To regulate the output voltage	U	CO2
5	What does a capacitor filter do in a power supply? A) Increase the voltage B) Smooth the output voltage C) Decrease the current D) Store electrical energy	U	C02
6	What is the primary function of a 78XX voltage regulator IC A) To increase voltage B) To provide a constant output voltage C) To store electrical energy D) To convert AC to DC	U	CO2
7	The output waveform of a bridge rectifier is A) A pure sinusoidal waveform B) A smooth DC signal C) A pulsating DC signal D) An inverted AC waveform	K	CO2

8	In a Brushless DC motor, the rotor contains which of the following? A) Permanent magnets B) Induced electromagnets C) Copper windings D) Brushes	U	CO3
9	In an induction motor, what is the relationship between the speed of the rotor and the synchronous speed? A) The rotor speed is always higher than the synchronous speed B) The rotor speed is always equal to the synchronous speed C) The rotor speed is always lower than the synchronous speed D) The rotor speed is independent of synchronous speed	A	CO3
10	What type of current is delivered to the battery in DC fast charging? A) Alternating Current (AC) B) Direct Current (DC) C) Pulsating Current D) Inductive Current	K	CO3

1 * 10 = 10

Part B

Short Answer Questions

Answer any**4** Questions

Each question carries 5 marks

11	What is the gain of an inverting amplifier ifR_f=10K and R_{in} =1K	U	CO1
12	What is an LDR. Explain with proper circuit diagram to amplify the signal from LDR using op-amp	U	CO1
13	Draw the output waveform of a Schmitt trigger whenV_{UT} =2V and V_{LT}= - 2V	U	CO2
14	What is a bridge rectifier.How does it convert AC voltage to DC voltage using diodes?	U	CO2
15	What is a Battery Management Unit (BMU)?	A	CO3
16	What is commutation. How is it achieved in a BLDC motor	A	CO3

4 * 5 = 20

Part C

Essay Questions

Answer any**2** Questions

Each question carries 10 marks

17	Explain the working of a non-inverting amplifier with proper circuit diagram. Write the gain formula of non-inverting amplifier	K	CO1
18	How we can use an IC 741 as a zero-crossing detector. Explain with proper circuit diagram and also draw the input and output waveforms	U	CO2
19	With the help of block diagrams explain the working of SMPS	U	CO2
20	Explain the working of BLDC motor. Why it is called brushless	A	CO3

2 * 10 = 20